

EXPENSE TRACKING SYSTEM
AI23431- WEBTECHNOLOGY AND MOBILE APPLICATION
MINI PROJECT REPORT

Submitted by

- ARUNTHIRAVIAM M (241501023)
-ATCHAYA R (241501026)

in partial fulfillment of the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE MACHINE LEARNING



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

An Autonomous Institute

CHENNAI - 602 105

MAY 2026

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled **“EXPENSE TRACKING SYSTEM”** is the Bonafide work of ARUNTHIRAVIAM M (2116241501023), ATCHAYA R(2116241501026), who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Dr. A RAJASEKARAN

Professor,

Department of Artificial Intelligence
and Machine Learning, Rajalakshmi
Engineering College,
Chennai - 602105.

Submitted to Project Viva-Voce Examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. K. SEKAR, Ph.D.**, Professor and Head of the Department of Artificial Intelligence and Machine Learning for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Mr. A.K.RAJASEKARAN**, Department of Artificial Intelligence and Machine Learning for his useful tips during our review to build our project.

ARUNTHIRAVIAM M-241501023

ATCHAYA R-241501026

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Machine Learning Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open-ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas,

inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVE

- To identify and formulate real-world problems that can be solved using Artificial Intelligence and Machine Learning techniques.
- To apply theoretical and practical knowledge of AI/ML for designing innovative, data-driven solutions.
- To integrate various tools, frameworks, and algorithms to develop, test, and validate AI/ML models.
- To demonstrate effective teamwork, project management, and communication skills through collaborative project execution.
- To instill awareness of ethical, societal, and environmental considerations in the design and deployment of intelligent systems.

COURSE OUTCOME

- **CO 1:** Analyze and define a real-world problem by identifying key challenges, project requirements and constraints.
- **CO 2:** Conduct a thorough literature review to evaluate existing solutions, identify research gaps and formulate research questions.

- **CO 3:** Develop a detailed project plan by defining objectives, setting timelines, and identifying key deliverables to guide the implementation process.
- **CO 4:** Design and implement a prototype or initial model based on the proposed solution framework using appropriate AI tools and technologies.
- **CO 5 :** Demonstrate teamwork, communication, and project management skills by preparing and presenting a well-structured project proposal and initial implementation results
- **CO - PO — PSO matrices of course**

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
AI23431.01	3	3	3	3	3	3	2	2	3	2	1	3	3	3	2
AI23431.02	3	3	3	3	3	3	2	2	3	2	1	1	3	3	2
AI23431.03	3	3	3	3	3	3	2	2	3	2	2	2	3	3	3
AI23431.04	3	3	3	3	3	3	2	2	3	2	2	3	3	3	3
AI23431.05	3	3	3	3	3	3	2	2	3	2	3	3	3	3	3
Average	3	3	3	3	3	3	2	2	3	2	1.8	2.1	3	3	2.4

- Note: Enter correlation levels 1, 2 or 3 as defined below:
- 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put “-”

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	Quality Education	The course provides fundamental knowledge in web and mobile application development, enabling students to gain practical technical skills, improve problem-solving abilities, and become industry-ready professionals.
SDG-8	Decent Work and Economic Growth	The subject equips students with job-oriented skills such as web development, client-side and server-side scripting, and mobile application design, increasing employability and supporting economic growth.
SDG-9	Industry, Innovation, and Infrastructure	The course promotes innovation by teaching modern technologies used in developing web and mobile applications, contributing to digital infrastructure and technological advancement.
SDG-10	Reduced Inequalities	By enabling the development of accessible web and mobile applications, the course helps bridge the digital divide and ensures equal access to digital services for diverse users.

ABSTRACT

SpendLens is an AI-powered expense tracking application designed to help users record, analyze, and understand their personal spending patterns in an efficient and user-friendly manner. With the growing challenge of unmanaged expenses and lack of financial awareness, the system aims to replace manual expense tracking methods with a smart, automated, and insightful digital platform. The application enables users to securely log in, record daily expenses, and visualize their spending behavior through interactive dashboards and charts.

The system provides secure user authentication with login and registration features to ensure data privacy. New users can create accounts by providing essential personal details, while existing users can access their personalized dashboards after verification. Once logged in, users can add expenses by selecting dates from a calendar, entering multiple expense breakdowns with corresponding amounts, and saving them for future reference. The application automatically calculates daily totals and displays them directly on the calendar for easy tracking.

SpendLens offers advanced data visualization through pie charts that categorize expenses and present insights based on different time periods such as today, weekly, monthly, year-to-date, and custom date ranges. These visual analytics help users clearly understand where their money is being spent and identify areas for potential savings. The dashboard also includes profile management and logout functionality for secure session handling.

The architecture of SpendLens is designed to be lightweight, responsive, and suitable for local or standard deployment environments. By combining structured expense recording with AI-driven insights and intuitive visualizations the system enhances financial awareness, encourages better budgeting habits, and supports informed decision-making. Overall, SpendLens serves as an effective personal finance management tool that simplifies expense tracking while promoting smarter spending behavior.

SI NO	TITLE	PAGE NO
	ACKNOWLEDGEMENT	iii
	ABSTRACT	xi
	LIST OF ABBREVIATIONS	xiv
1	INTRODUCTION	1
1.1	General	1
1.2	Need for Expense Tracking System	2
1.3	Scope of the System	3
1.4	Organization of the Report	3
2	LITERATURE SURVEY	4
2.1	Introduction	4
2.2	Existing Expense Tacking System	4
2.3	Related Work	5
2.4	Summary	5
3	PROBLEM FORMULATION AND OBJECTIVES	6
3.1	Problem Statement	6
3.2	Objectives of the System	6
4	SYSTEM DESIGN	7
4.1	Introduction	7
4.2	System Architecture	7
4.3	Database Design	7

4.4 System Requirements	8
4.4.1 Hardware Requirements	8
4.4.2 Software Requirements	8
5 SYSTEM IMPLEMENTATION	8
5.1 System Methodology	8
5.2 Module Descriptions	9
Index Module	9
5.2.1 Login Module	10
5.2.2 Registration Module	12
5.2.3 Dashboard Module	13
5.2.4 Expense Entry Module	14
6 RESULTS AND DISCUSSION	15
6.1 Result	15
6.2 Performance Evaluation	16
6.3 Resource Utilization	17
7 CONCLUSION AND FUTURE WORK	17
7.1 Conclusion	17
7.2 Future Work	18
REFERENCES	18
APPENDICES	19
Source Code	19
Conclusion	25

LIST OF ABBREVIATIONS

DB	– Database
SQL	– Structured Query Language
API	– Application Programming Interface
UI	– User Interface
HTML	– HyperText Markup Language
JS	– JavaScript
HTTP	– HyperText Transfer Protocol
SMTP	– Simple Mail Transfer Protocol
CRUD	– Create, Read, Update, Delete

CHAPTER 1

INTRODUCTION

1.1 General

Personal expense management plays a crucial role in maintaining financial stability and improving money-handling habits. Many individuals fail to track their daily expenses properly, leading to overspending and lack of awareness about where their money is being spent. Traditional methods such as maintaining notebooks or manual spreadsheets are time-consuming, error-prone, and often neglected due to inconvenience. These methods also make it difficult to analyze spending patterns or generate meaningful insights.

SpendLens is designed to automate and simplify the process of tracking and managing personal expenses through a digital and intelligent platform. The system allows users to securely register and log in, record daily expenses by date, and categorize spending in an organized manner. By storing expense data in a centralized database, the application ensures accuracy, easy retrieval of records, and efficient financial tracking.

The system provides an interactive dashboard with calendar-based expense entry and visual representations such as pie charts to help users understand their spending behavior. Users can view expenses for specific time periods including daily, weekly, monthly, year-to-date, and custom date ranges. This structured approach helps users identify spending trends, control unnecessary expenses, and plan savings effectively.

By reducing the manual effort involved in expense tracking and offering AI-based insights and visual analytics, SpendLens enhances financial awareness and supports informed decision-making. The application serves as a user-friendly and

reliable solution for personal finance management, promoting disciplined spending habits and better financial control.

1.2 Need for Expense Tracking System

Many individuals still rely on traditional methods such as handwritten notes or basic spreadsheets to track their expenses, which can lead to several challenges including inaccurate records, missed entries, and lack of financial clarity. Manual expense tracking requires consistent effort and discipline, and users often fail to maintain long-term records or analyze their spending patterns effectively.

A digital expense tracking and management system helps overcome these limitations by providing an automated and organized solution. SpendLens enables users to record expenses quickly, store data securely in a centralized database, and access their financial information anytime. By categorizing expenses and associating them with specific dates, the system simplifies expense monitoring and improves accuracy.

The application enhances financial transparency by allowing users to view detailed breakdowns of their spending and visualize expenses through charts and dashboards. It reduces manual effort, minimizes human errors, and supports better budgeting decisions. Additionally, the system helps users identify overspending patterns and receive AI-driven insights, enabling improved financial planning and smarter money management.

1.3 Scope of the System

The SpendLens system is developed to simplify the process of recording, storing, and analyzing personal expense data in an efficient and user-friendly manner. The system provides secure access to users through login and registration authentication, ensuring that financial information is protected and accessible only to authorized individuals.

Users can add daily expenses by selecting dates from a calendar and entering multiple expense breakdowns along with corresponding amounts. The system stores all expense records in a centralized database, allowing users to retrieve and manage their financial data easily. SpendLens also provides an interactive dashboard that displays expense summaries and visual insights using pie charts for various time periods such as daily, weekly, monthly, year-to-date, and custom date ranges.

The scope of the system includes improving accuracy in expense tracking, reducing manual effort, and enhancing financial awareness through clear visualizations and AI-driven insights. The application can be used by individuals for personal finance management and budgeting purposes. SpendLens is suitable for deployment on standard systems and can be extended in the future to include advanced analytics, budgeting alerts, and integration with banking platforms.

1.4 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the **SpendLens – Expense Tracker with AI Insights** system.

Chapter 1: Provides an introduction to the project, including the general overview, need for the expense tracking system, and scope of the project.

Chapter 2: Presents the literature survey, which discusses existing expense tracking applications, personal finance management systems, and related work.

Chapter 3: Explains the problem formulation and objectives of the proposed SpendLens system.

Chapter 4: Describes the system design, including the overall architecture, database design, and system requirements.

Chapter 5: Discusses the implementation of the system and explains the various modules such as user authentication, expense entry, dashboard, calendar view, and data visualization.

Chapter 6: Presents the results and performance evaluation of the system, including usability and accuracy of expense analysis.

Chapter 7: Concludes the project and discusses possible future enhancements such as advanced AI recommendations, budgeting alerts, and integration with financial services.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Expense tracking is an important activity for individuals as it helps in monitoring spending habits and maintaining financial discipline. Traditionally, expenses are recorded manually using notebooks or simple spreadsheets, which can be time-consuming and prone to errors. These manual methods also make it difficult to analyze spending patterns and manage long-term financial records.

With the advancement of technology, various digital expense tracking systems have been developed to improve efficiency and accuracy. Modern applications use web-based and mobile platforms to record expenses automatically and store data securely. Recent studies highlight the use of data analytics and AI-based techniques to categorize expenses and provide meaningful financial insights, helping users understand their spending behavior more effectively.

2.2 Existing Expense Tracking Systems

Many individuals and organizations use digital expense tracking applications to replace traditional manual methods. Some systems allow users to enter expenses manually by selecting categories such as food, travel, and utilities. These applications help users organize expenses and reduce the effort required to maintain records.

Other systems integrate with banking services to automatically fetch transaction details and classify expenses. These systems analyze transaction data and display spending summaries using charts and reports. While such applications improve convenience and accuracy, they often require internet access and user permission to access financial data.

2.3 Related Work

Several research studies have proposed different approaches to automate personal expense tracking and financial management. A systematic review of expense tracking and budgeting applications shows that digital systems significantly improve accuracy and consistency when compared to manual methods. These systems also help users maintain centralized financial records and generate analytical reports to understand spending behavior.

Other studies focus on integrating technologies such as data analytics, artificial intelligence, and cloud-based storage to provide real-time expense monitoring and personalized financial insights. These solutions reduce manual effort and enhance reliability in maintaining expense records while supporting better budgeting and financial planning.

2.4 Summary

From the literature survey, it is evident that digital expense tracking systems offer significant advantages over traditional manual methods. They improve accuracy, reduce time consumption, and enable centralized data management. However, many existing systems depend on bank integrations or continuous internet access, which may limit usability for some users. Therefore, a web-based expense tracking system like SpendLens provides a simple and cost-effective solution that allows users to manage expenses digitally while offering clear visual insights into their spending patterns.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

Many individuals still track their daily expenses manually using notebooks or simple spreadsheets. This traditional approach is time-consuming and often leads to issues such as missed entries, inaccurate calculations, and difficulty in maintaining long-term expense records. Users also find it challenging to analyze spending patterns and calculate total expenses over specific time periods.

Users often lack a clear understanding of where their money is being spent, which can result in poor financial planning and overspending. Manual methods do not provide visual insights or categorized summaries, making it difficult for users to monitor their financial habits effectively. Additionally, managing and reviewing large amounts of expense data manually reduces efficiency.

Therefore, there is a need for an automated system that can simplify the process of recording, storing, and analyzing personal expenses. A digital expense tracking system like SpendLens can help maintain accurate financial records, provide quick access to expense data, and offer meaningful insights to support better financial decision-making.

3.2 Objectives of the System

The main objective of the SpendLens system is to develop a digital platform that simplifies the process of tracking and managing personal expenses. The system aims to provide an efficient and reliable way for users to record expenses and monitor their spending behavior.

The specific objectives of the system include:

- To maintain expense records in a centralized database.
- To provide secure login and registration for users.
- To allow users to view expense details based on date and category.
- To reduce manual effort and errors in expense tracking.
- To improve financial awareness through visual insights and summaries.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system helps improve efficiency, reliability, and maintainability.

SpendLens is designed as a web-based application that allows users to record, manage, and analyze personal expenses. The system follows a client-server architecture where users interact through a web interface, while the backend handles data processing, business logic, and storage in a centralized database.

4.2 System Architecture

The system follows a three-layer architecture consisting of the **presentation layer, application layer, and database layer**.

•PresentationLayer:

This layer represents the user interface where users interact with the system. It includes web pages such as login and registration pages, dashboard, calendar view, expense entry forms, and data visualization pages developed using HTML, CSS, and JavaScript.

•ApplicationLayer:

This layer handles the business logic of the system. It processes user requests, verifies login credentials, manages expense entries, calculates totals, and generates analytical insights. The backend of the system is implemented using Node.js.

•DatabaseLayer:

This layer stores all system data including user details, expense breakdowns, dates, and amount records. The database ensures secure storage and easy retrieval of expense information whenever required.

This architecture ensures smooth communication between users and the database while maintaining system efficiency, scalability, and reliability.

4.3 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

4.3.1 Hardware Requirements

The SpendLens system does not require high-end hardware and can run on standard computers.

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 GB Hard Disk or SSD
- Input Devices: Keyboard and Mouse
- Internet Connection

4.3.2 Software Requirements

The system is developed using commonly available software tools and technologies.

- Operating System: Windows 10 or later
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Technology: Node.js
- Database: MySQL
- Development Environment: Visual Studio Code
- Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

SpendLens is implemented using a web-based approach that allows users to access the system through a browser. The development process involves designing the user interface, implementing backend functionalities, and integrating the database for storing expense records.

The frontend of the system is developed using HTML, CSS, and JavaScript, which provide a simple and interactive interface for users to record and view their expenses. The backend is implemented using Node.js, which handles user authentication, expense processing, calculations, and communication with the database. The system uses a MySQL database to store user information, expense breakdowns, dates, and amount details.

The system follows a modular approach where each functionality such as login, registration, expense entry, dashboard, and data visualization is developed as a separate module. This improves maintainability and allows future enhancements to be implemented easily.

5.2 Module Descriptions

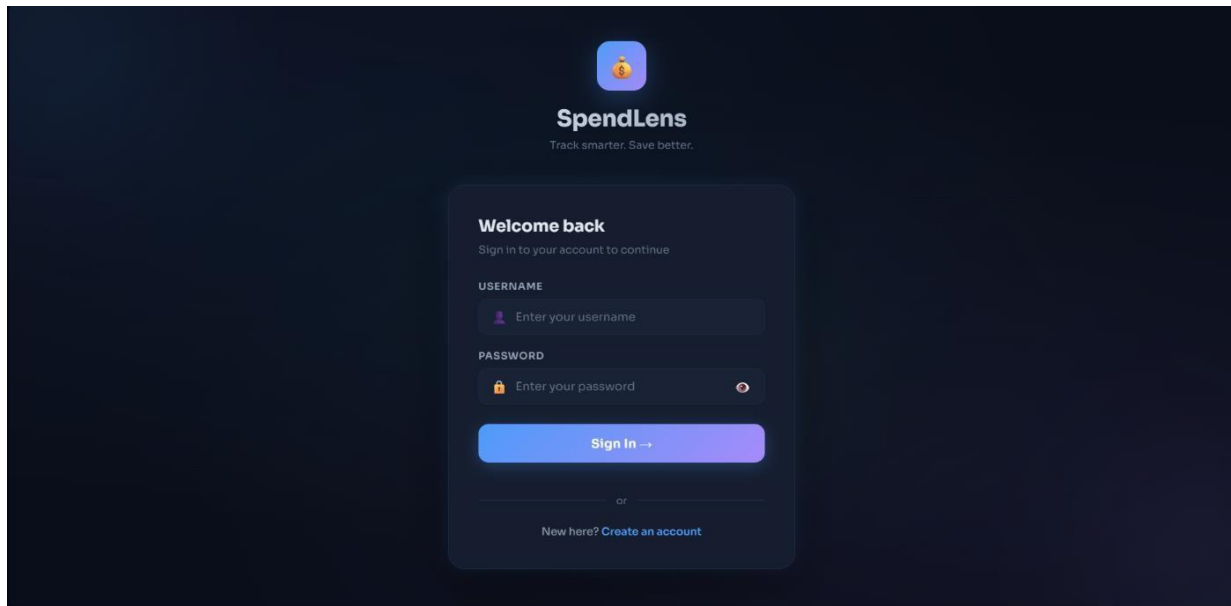
The SpendLens system is composed of multiple functional modules that work together to provide a secure, efficient, and user-friendly expense tracking and analysis platform. Each module is designed to perform a specific task while ensuring smooth interaction with other modules of the system.

Index Module

The Index Module acts as the initial landing page of the SpendLens application. It introduces users to the system and provides a clear overview of its purpose and features before authentication. The page displays the application name *SpendLens* along with the tagline that highlights smart expense tracking and better savings.

The index page showcases key features such as secure authentication, expense categorization, calendar-based tracking, visual analytics, and AI-powered insights. These features help users understand the capabilities of the system at a glance. The module provides navigation options for users to proceed to the login page or the registration page.

This module plays an important role in improving user reminder and ensuring easy navigation. The simple and visually appealing design enhances the first-time user experience and encourages users to explore the application further.

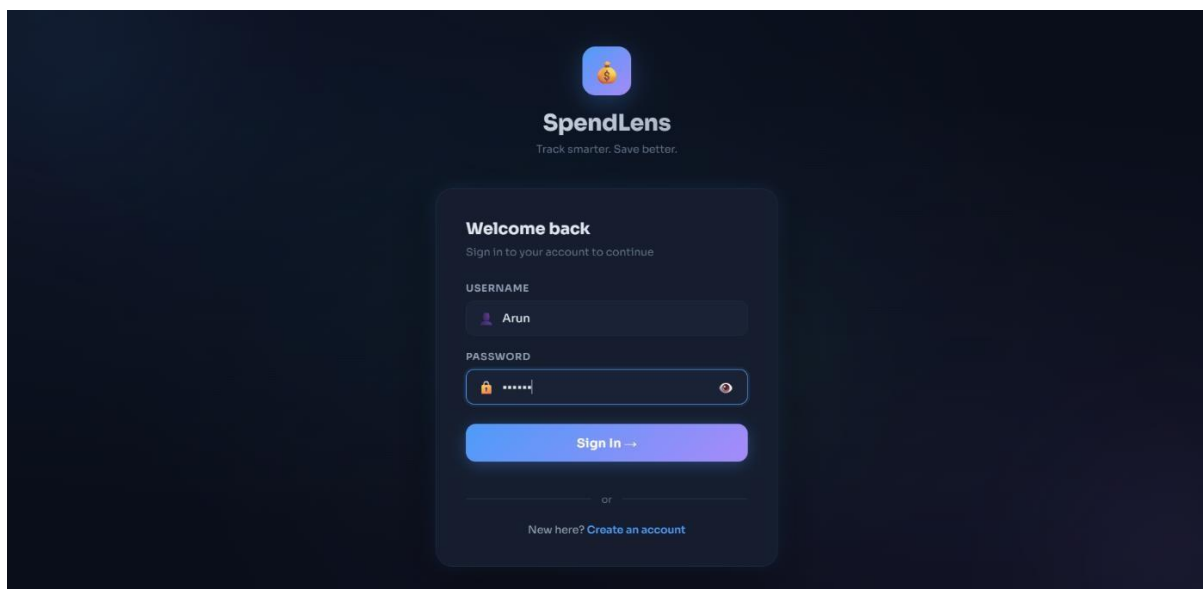
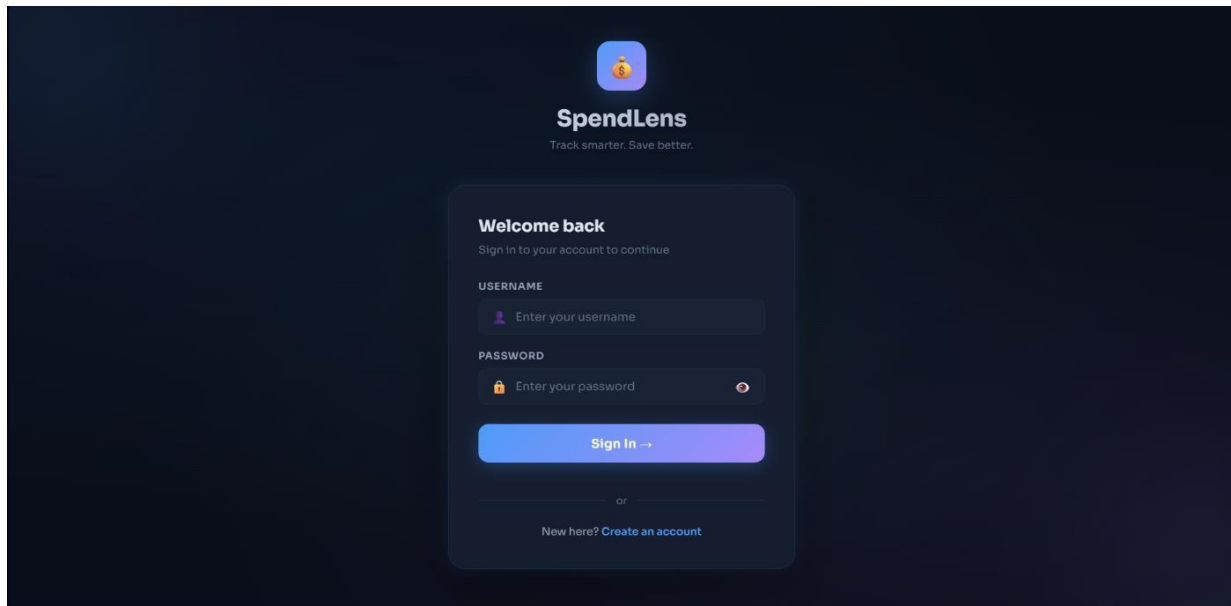


5.2.1 Login Module

The Login Module is responsible for authenticating existing users and granting secure access to the system. Users are required to enter their username and password to log in. The entered credentials are verified with the records stored in the database.

If the credentials are valid, the system allows the user to access the dashboard. If incorrect credentials are entered, appropriate error messages are displayed to guide the user. The login page also includes a “Create Account” link that redirects new users to the registration module.

This module ensures data security by restricting access to authorized users only. It also manages session handling, ensuring that user data remains protected throughout the usage of the application.

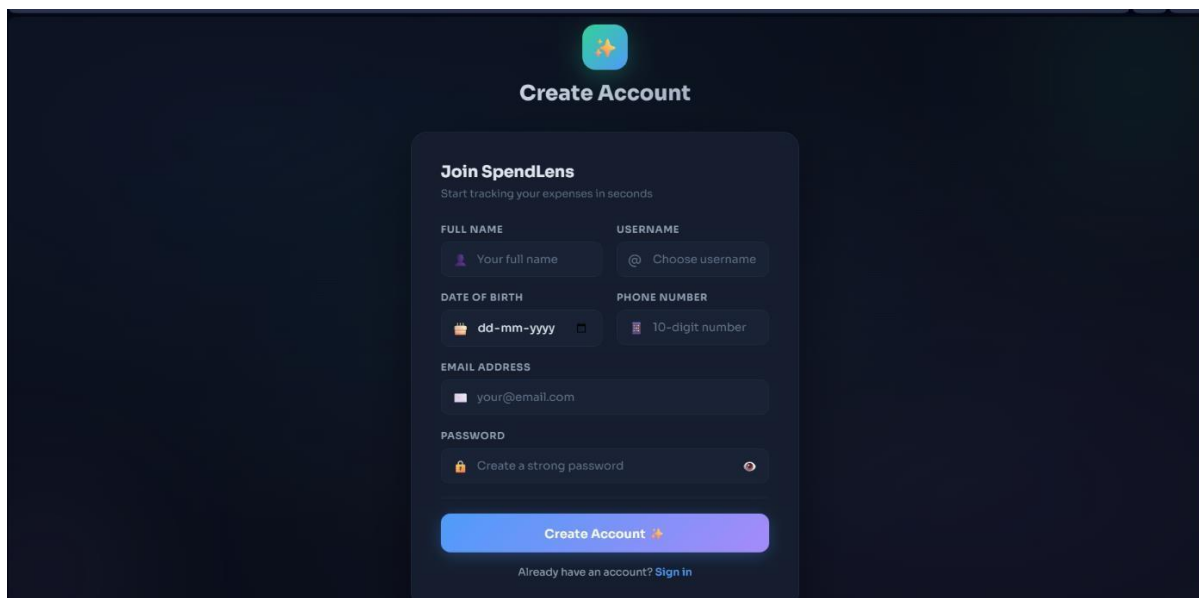


5.2.2 Registration Module

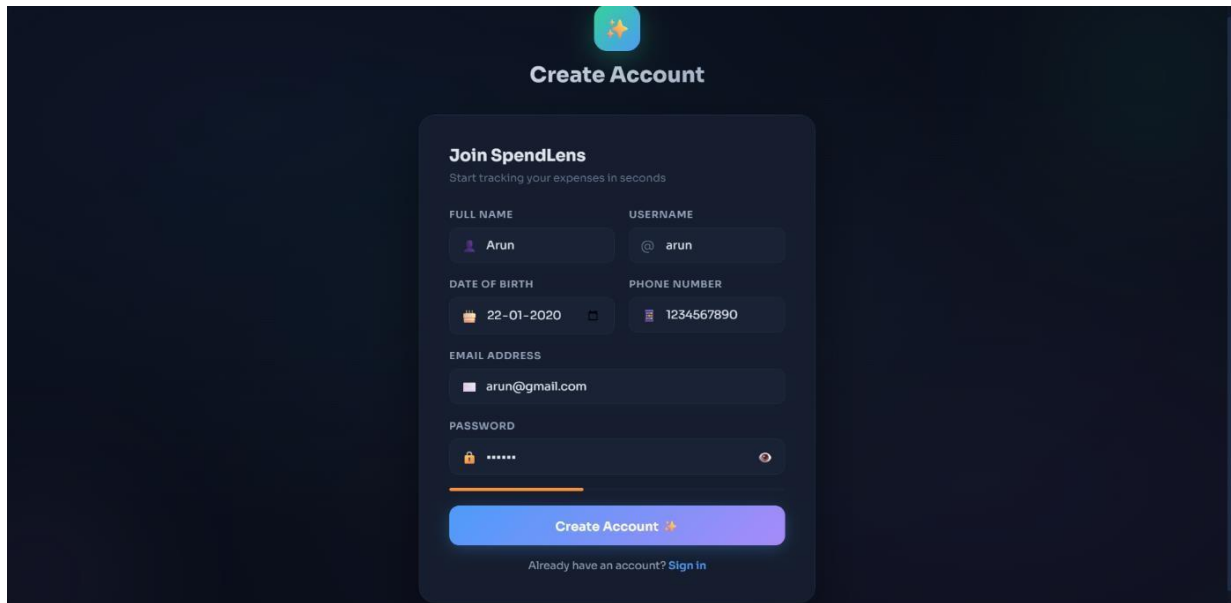
The Registration Module allows new users to create an account in the SpendLens system. The registration form collects essential user information such as full name, username, date of birth, phone number, email address, and password.

The system validates the entered details to ensure correctness and completeness. Once the validation is successful, the user information is securely stored in the database. Passwords are handled securely to protect user privacy.

After successful registration, the user is redirected to the login page, where they can log in using their newly created credentials. This module ensures smooth onboarding of new users and prevents duplicate or invalid account creation.



The screenshot displays a dark-themed user interface for the 'Create Account' page. At the top center, there is a logo consisting of three colored dots (green, yellow, and blue) above the text 'Create Account'. Below this, a central card titled 'Join SpendLens' contains the registration form. The card has a subtitle 'Start tracking your expenses in seconds'. The form includes several input fields: 'FULL NAME' with a placeholder 'Your full name', 'USERNAME' with a placeholder '@ Choose username', 'DATE OF BIRTH' with a placeholder 'dd-mm-yyyy' and a calendar icon, 'PHONE NUMBER' with a placeholder '10-digit number' and a phone icon, 'EMAIL ADDRESS' with a placeholder 'your@email.com', and 'PASSWORD' with a placeholder 'Create a strong password' and a password strength indicator icon. A large blue button labeled 'Create Account' with a star icon is positioned below the fields. At the bottom of the card, a link reads 'Already have an account? Sign in'.



Create Account

Join SpendLens
Start tracking your expenses in seconds

FULL NAME: Arun

USERNAME: arun

DATE OF BIRTH: 22-01-2020

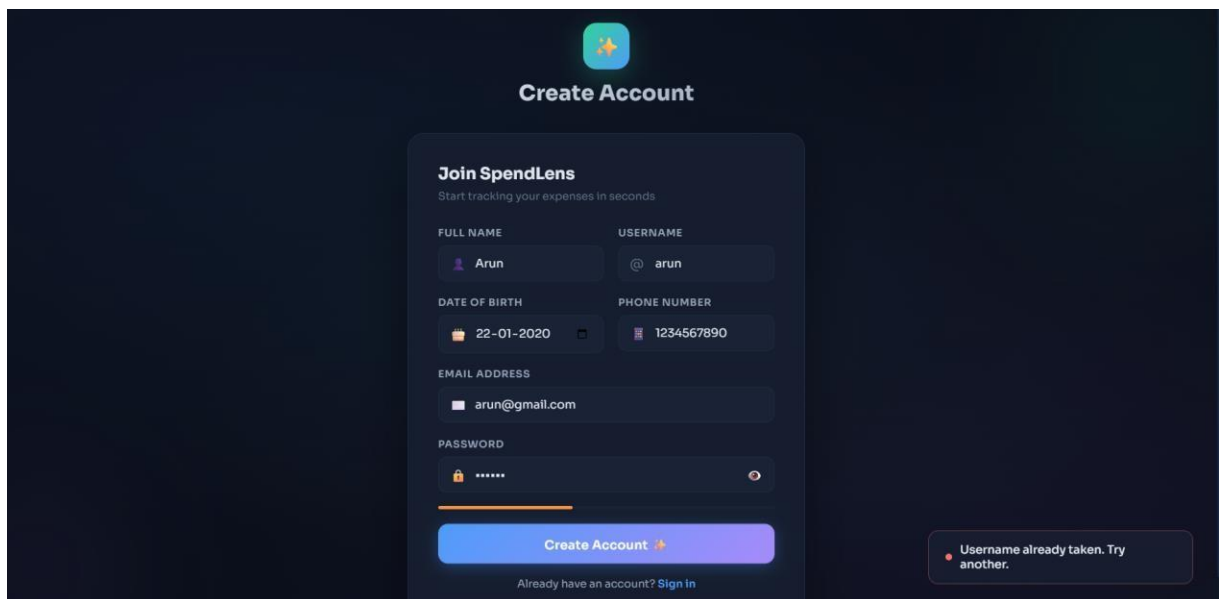
PHONE NUMBER: 1234567890

EMAIL ADDRESS: arun@gmail.com

PASSWORD: *****

Create Account

Already have an account? [Sign In](#)



Create Account

Join SpendLens
Start tracking your expenses in seconds

FULL NAME: Arun

USERNAME: arun

DATE OF BIRTH: 22-01-2020

PHONE NUMBER: 1234567890

EMAIL ADDRESS: arun@gmail.com

PASSWORD: *****

Create Account

Already have an account? [Sign In](#)

Username already taken. Try another.

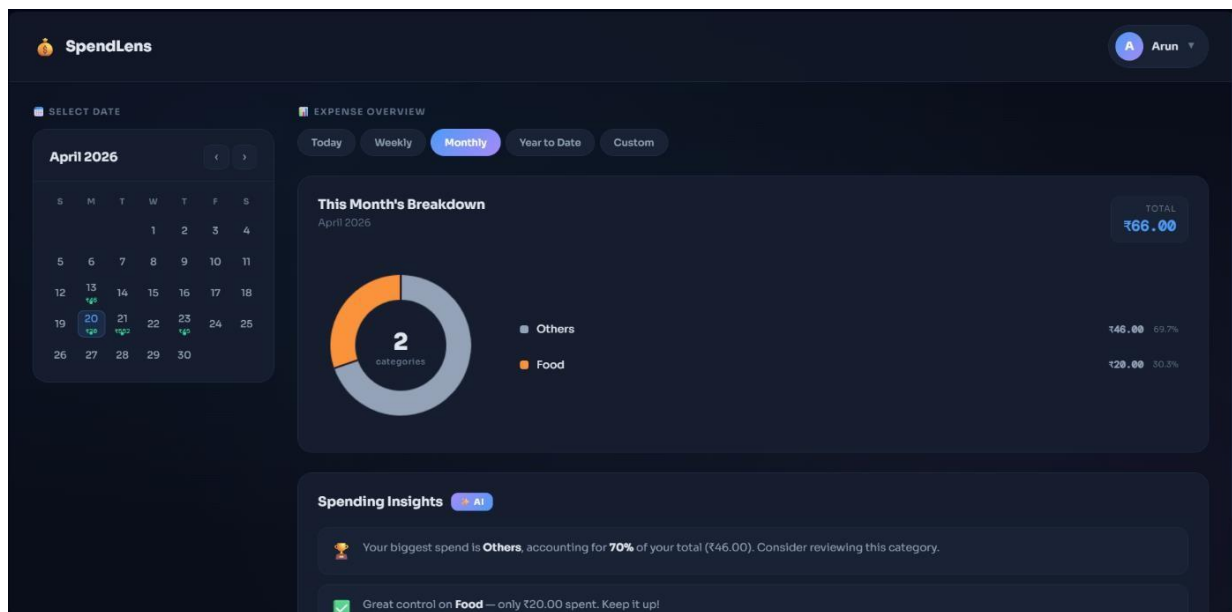
5.2.3– Dashboard Module

The Dashboard Module is the core interface of the SpendLens system and is displayed after successful login. It provides users with an overview of their expense data in a visually clear and organized manner. The dashboard contains a calendar that allows users to select any date to view or add expenses.

For each date, the total expense amount is displayed directly on the calendar, helping users quickly identify high or low spending days. The dashboard also includes a pie chart that visually represents expense distribution based on

selected time periods such as today, weekly, monthly, year-to-date, and custom date ranges.

A profile icon is available at the top-right corner of the dashboard, providing access to logout functionality. This module helps users analyze spending trends, monitor expenses efficiently, and gain better financial awareness.

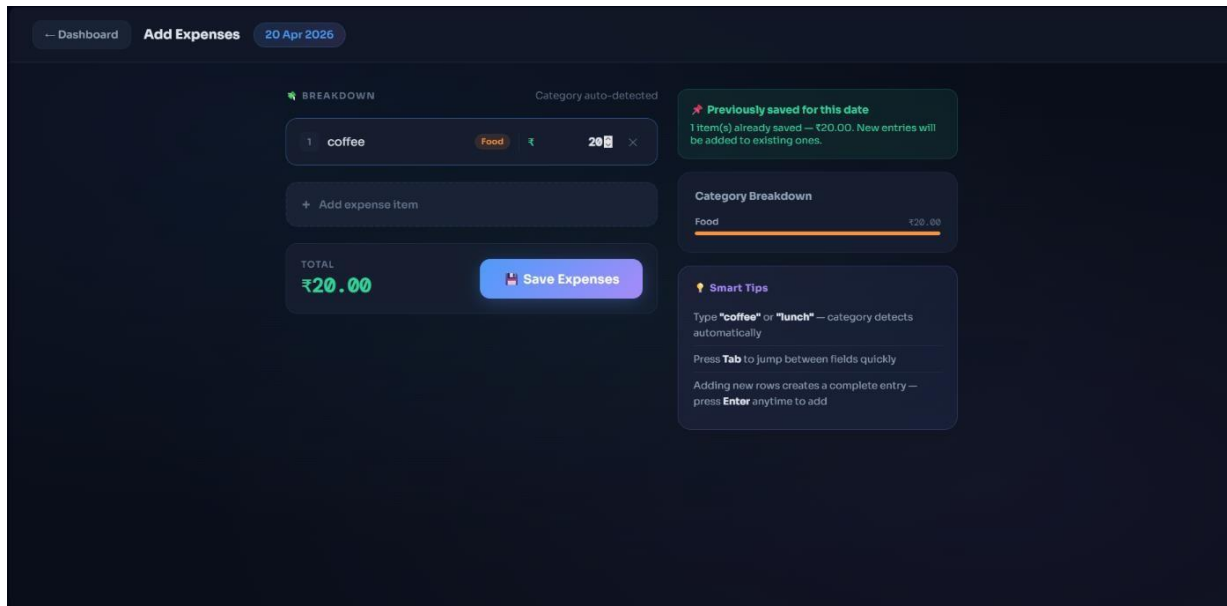


5.2.4 Expense Entry Module

The Expense Entry Module enables users to record daily expenses in a structured format. When a user selects a specific date from the calendar, the system opens a dedicated expense entry page for that date.

The page consists of two columns: one for expense breakdown and another for corresponding amounts. Users can add multiple expense entries using a “+” button, allowing flexible and detailed expense recording. Each expense entry is saved temporarily and finalized when the user clicks the save button.

After saving, the system calculates the total expense for the selected date and stores the data in the database. The user is then redirected back to the dashboard, where the updated total is displayed on the calendar. This module ensures accurate daily expense tracking and easy data management.



CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The SpendLens expense tracking system was successfully designed, implemented, and tested as a web-based application. The system enables users to record, store, and analyze their daily expenses through a secure and interactive platform. All major functionalities, including user authentication, expense entry, dashboard visualization, and data retrieval, were tested and found to be working efficiently.

During the testing phase, the system performed reliably in handling user registration and login operations. Users were able to create accounts, log in securely, and access their personalized dashboards without errors. Expense entries made for different dates were stored accurately in the database and retrieved correctly when required. The calendar-based interface allowed users to view daily expense totals clearly, improving ease of tracking.

The dashboard module successfully displayed expense distribution using pie charts for various time filters such as today, weekly, monthly, year-to-date, and custom date ranges. These visual representations helped users understand their spending patterns effectively. The system also ensured smooth navigation between modules, including redirection from expense entry pages back to the dashboard.

Overall, the implementation of SpendLens achieved its primary objective of simplifying personal expense management. The system reduced manual effort, improved accuracy in expense tracking, and provided meaningful insights through visual analytics, making it a reliable and efficient tool for personal financial management.

6.2 Performance Evaluation

The performance of the SpendLens system was evaluated based on usability, accuracy, and response time. The application provides a clean and user-friendly interface that allows users to navigate easily between modules such as login, dashboard, calendar view, and expense entry. The intuitive design reduces user effort and improves overall interaction with the system.

Expense data is stored in a centralized database, ensuring accurate record maintenance and fast retrieval whenever required. The use of Node.js for backend processing enables the system to handle user requests efficiently, including authentication, expense calculations, and dashboard updates. During testing, the system responded quickly to user actions such as saving expenses and switching between different time-based views.

The system performed reliably under normal usage conditions without noticeable delays or data inconsistencies. The modular architecture of SpendLens allows new features such as advanced AI insights, budgeting alerts, or report generation to be integrated in the future without affecting existing performance. Overall, the system demonstrated stable performance and met the expected functional and non-functional requirements.

6.3 Resource Utilization

The Student Attendance Management System is designed to operate efficiently on standard computer systems without the need for high-end hardware resources. The system makes use of lightweight and widely supported web technologies such as HTML, CSS, and JavaScript for the frontend, which ensures low memory consumption and smooth user interaction.

The backend of the system is implemented using Node.js, which efficiently handles server-side operations such as user authentication, attendance processing, and communication with the database. Node.js uses an event-driven and non-blocking architecture, allowing the system to manage multiple user requests with minimal CPU and memory usage.

The MySQL database is structured to store attendance records, student details, and faculty information in an organized manner. Efficient database queries ensure quick data retrieval while minimizing storage and processing overhead. During testing, the system demonstrated stable performance with low resource consumption, even on systems with basic hardware configurations.

Since the application is web-based, users can access the system through a standard web browser without installing any additional software. This reduces system overhead and makes the application cost-effective, resource-efficient, and easy to deploy in educational institutions.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The SpendLens system was developed to provide an efficient and reliable solution for managing personal expenses and improving financial awareness. The system replaces traditional manual expense tracking methods with a digital platform that simplifies the process of recording, storing, and analyzing expense information.

The application allows users to securely log in and record daily expenses using a calendar-based interface, while also enabling them to view detailed expense summaries and visual insights through an interactive dashboard. By maintaining

expense data in a centralized database, the system ensures accuracy, transparency, and quick retrieval of financial information whenever required.

The implementation of SpendLens successfully reduces manual effort, minimizes errors in expense tracking, and enhances overall efficiency in personal finance management. The web-based design makes the system accessible, user-friendly, and suitable for everyday use, helping users develop better spending habits and make informed financial decisions.

7.2 Future Work

Although the current SpendLens system provides essential features for personal expense tracking and analysis, several improvements can be implemented in the future to enhance its functionality. Future enhancements may include automatic expense detection through bank or UPI integration to reduce manual data entry.

Mobile application support can also be developed to allow users to track expenses conveniently through smartphones. Additional features such as advanced AI-driven insights, personalized budgeting recommendations, spending alerts, and predictive expense analysis can further improve the effectiveness of the system.

These enhancements will make SpendLens more intelligent, user-friendly, and suitable for a wider range of users, supporting better financial planning and long-term money management.

REFERENCES

- [1] A. Kapoor and S. Gupta, “A Study on Personal Expense Tracking and Budgeting Applications,” *International Journal of Computer Applications*, vol. 176, no. 32, pp. 15–20, 2021.**
- [2] R. Thakur and P. Jain, “Digital Financial Management Systems and Their Impact on Personal Budgeting,” *Journal of Information Systems and Technology*, vol. 9, no. 2, pp. 45–52, 2020.**
- [3] S. Rathi, M. Sharma, and K. Verma, “Analysis of Expense Tracking Systems Using Data Analytics,” *International Journal of Advanced Research in Computer Science*, vol. 11, no. 4, pp. 98–103, 2020.**

[4] Node.js Foundation, “Node.js Documentation,” Available: <https://nodejs.org>

[5] MySQL Documentation, Oracle Corporation, “MySQL Reference Manual,” Available: <https://dev.mysql.com/doc>

[6] T. Berners-Lee, “HyperText Markup Language (HTML),” World Wide Web Consortium (W3C), Available: <https://www.w3.org>

[7] D. Flanagan, *JavaScript: The Definitive Guide*, 7th ed., O’Reilly Media, 2020.

APPENDICES

Appendix – 1 : System Overview & Module Description

System Overview

SpendLens is a smart personal finance management system designed to help users track expenses, analyze spending patterns, and improve financial discipline. The system allows users to record daily expenses, categorize transactions, visualize spending insights, and receive alerts when budgets are exceeded.

SpendLens focuses on:

- Expense tracking and categorization
- Monthly and category-wise analysis
- Budget monitoring and alerts
- Simple and user-friendly interface

The system can be implemented using Node.js / Python, MySQL, and a web or mobile-based frontend.

Module 1: System Initialization Module

Purpose

To initialize the application, database connection, and user interface.

Pseudocode

START

Connect to database

IF connection successful THEN

 Start server

 Load SpendLens homepage

ELSE

 Display database error

END IF

END

Module 2: User Authentication Module

Purpose

To provide secure authentication for users.

Pseudocode

FUNCTION Login(email, password)

Check user table for matching credentials

IF valid THEN

 Create user session

 Redirect to dashboard

ELSE

 Display "Invalid login details"

END IF

END FUNCTION

Module 3: Expense Entry Module

Purpose

To allow users to add daily expenses with category and amount.

Pseudocode

FUNCTION AddExpense(user_id, amount, category, date, note)

Insert expense into expense table

Return "Expense added successfully"

END FUNCTION

Module 4: Expense Categorization Module

Purpose

To classify expenses into categories like Food, Travel, Rent, Shopping, etc.

Logic

Categories:

Food

Transport

Rent

Entertainment

Utilities

Others

Each expense is linked to one category

Module 5: Expense Analysis & Visualization Module

Purpose

To analyze spending patterns and show insights.

Pseudocode

FUNCTION AnalyzeExpenses(user_id)

Retrieve expenses grouped by category and month

Calculate:

Total spending

Category-wise spending

Monthly spending trend

Display charts and summaries

END FUNCTION

Module 6: Budget Management Module

Purpose

To allow users to set monthly budgets and track limits.

Pseudocode

FUNCTION SetBudget(user_id, category, limit)

Store budget limit in budget table

END FUNCTION

Module 7: Budget Alert & Notification Module

Purpose

To notify users when spending exceeds budget.

Pseudocode

FUNCTION CheckBudget(user_id, category)

IF total_spent > budget_limit THEN

 Send alert notification

END IF

END FUNCTION

Module 8: Expense History & Report Module

Purpose

To allow users to view past expenses and download reports.

Pseudocode

FUNCTION ViewExpenseHistory(user_id)

 Retrieve all expense records

 Display date, amount, category, note

END FUNCTION

Module 9: Data Security & Validation Module

Purpose

To ensure secure and accurate data storage.

Features

- Input validation
- Secure password storage

- Session-based authentication
- User-specific data access

System Features Implemented

- Secure user login
- Daily expense tracking
- Category-wise expense management
- Budget setting and alerts
- Monthly spending analysis
- Visual charts and summaries
- Simple and responsive UI

Appendix 2

System Test Verification

Test Case Results

Test Case	Description	Result
Test Case 1	User Login	PASS
Test Case 2	Expense Entry	PASS
Test Case 3	Expense Categorization	PASS
Test Case 4	Budget Alert System	PASS
Test Case 5	Expense Report Viewing	PASS

Final System Status

The **SpendLens** system has been successfully designed and tested.

It provides an efficient way for users to track expenses, control spending habits, and gain financial insights.

With scalable architecture and user-friendly design, SpendLens is suitable for personal finance management and future enhancements.

CONCLUSION

The **SpendLens** system was developed to provide a simple, efficient, and user-friendly solution for personal expense tracking and financial management. The application helps users record daily expenses, categorize spending, and analyze their financial habits through a structured and organized platform. By replacing traditional manual expense tracking methods, SpendLens makes financial monitoring easier and more accurate.

The system allows users to add expenses with details such as amount, category, and date, and view their spending history through an interactive dashboard. Budget management and alert features help users control unnecessary expenses and maintain financial discipline. The centralized database ensures secure storage and quick retrieval of expense data whenever required.

The implementation of SpendLens successfully reduces manual effort, minimizes calculation errors, and improves awareness of spending patterns. The web-based design ensures accessibility across devices without requiring additional software installation. Overall, SpendLens achieves its objective of simplifying expense management and supporting better financial decision-making for users.